

Khulna University
Science, Engineering and Technology School
Discipline: **Mathematics**, Program: **B.Sc. (Hons.)**
Year: **Fourth**, Term: **I** Examination-2026; Session: **2024-2025**
Course Code: **0541 12 Math 4101**
Course Title: **Hydrodynamics and Hydrostatics**
Total Marks: **60**, Time: **03** Hours, Credit: **03**



- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions **Marks**

- | | | | |
|----|-----|--|----|
| 1. | (a) | Define stream tube, stream filament, rotational and irrotational flow. Distinguish between stream lines and path lines. [L4; CLO1] | 05 |
| | (b) | Derive a relation between local and individual rates of changes. Also briefly discuss about velocity potential or velocity function. [L5; CLO1] | 05 |
| 2. | (a) | Given the velocity components: $u = \frac{ax-by}{x^2+y^2}, v = \frac{ay+bx}{x^2+y^2}, w = 0$. Explain these components for a possible fluid flow. Also test its rotationality. [L4; CLO2] | 04 |
| | (b) | Deduce the equation of conservation of mass of fluid. If $q = -i(3y^2) - j(6x)$, determine the stream lines at (1, 1). [L5; CLO1] | 06 |
| 3. | (a) | State and prove Bernoulli's theorem for steady motion of an inviscid fluid. [L5; CLO2] | 05 |
| | (b) | Derive Euler's equation of motion in the form $\frac{\partial q}{\partial t} + \frac{1}{2} \nabla q^2 - q \times \text{curl } q = F - \frac{1}{\rho} \nabla p$. [L5; CLO2] | 05 |
| 4. | (a) | Find the expression to the complex potential for the streaming motion past an ellipse in terms of the elliptic co-ordinates and hence find the stream function. [L6; CLO3] | 05 |
| | (b) | Prove that the velocity function $\psi = U(r + \frac{a^2}{r}) \cos \theta$ represents a streaming motion past a fixed circular cylinder. [L6; CLO3] | 05 |

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

5. (a) Define source and sink. What are the values of velocity, stream function and velocity potential at the sink and source. Find the stream function if a source is in a uniform stream. [L1; CLO3] **05**
- (b) What is doublet? Derive the complex potential due to a doublet. [L5; CLO2,3] **05**
6. (a) State and prove Blasius theorem. [L5; CLO3] **05**
- (b) Discuss the motion represented by $W = \frac{1}{2} U \frac{a^3}{z^2}$, and show that the stream lines are lemniscates. [L5; CLO3] **05**
7. (a) Find the strength of a circular vortex. Show that the circulation can be considered as the measure of velocity potential around a closed circuit. [L2; CLO4] **05**
- (b) Calculate the circulation of the velocity field $\vec{q} = \frac{-yi + xj}{x^2 + y^2}$ round a square with corners (1, 0), (2, 0), (2, 1) and (1, 1). [L4; CLO4] **05**
8. (a) Find the center of pressure of a triangular area, the depth of whose vertices are α, β, γ from the free surface. [L5; CLO4] **05**
- (b) A regular hexagon is immersed in a liquid with one side in the surface. Show that the depth of its centre of pressure is to that of its centre of gravity as 23:18. [L6; CLO4] **05**

***** THE END *****

Khulna University
Science, Engineering and Technology School
Discipline: Mathematics, Program: B.Sc. (Hons.)
Year: Fourth, Term: I Examination-2026; Session: 2024-2025
Course Code: 0541 12 Math 4105
Course Title: Topology
Total Marks: 60, Time: 03 Hours, Credit: 03



- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

	Marks
1. (a) Define topological space. Also give example of usual topology, discrete and indiscrete topology. [L1; CLO1]	03
(b) Let $X = \{a, b, c, d, e\}$, write three subclasses of X which will form topology. Check whether it is a topology or not? Find the derived set of $A = \{a, b, c\}$ with respect to the topology T . [L1; CLO2]	03
(c) Define interior, exterior and boundary point. Let $T = \{X, \phi, \{a\}, \{c, d\}, \{a, c, d\}, \{b, c, d, e\}\}$ be a topology on $X = \{a, b, c, d, e\}$. Find the interior, exterior and boundary point of $A = \{a, b, c\}$. [L2; CLO3]	04
2. (a) Define limit point. Find the limit point of set of rational numbers. [L2; CLO3]	02
(b) State and prove Minkowski's inequality. [L3; CLO4]	04
(c) Define normed space. Let $I = [0, 1]$, $\ f\ = \sup\{ f(x) \}$. Prove that it is a norm on $C[0, 1]$. [L3; CLO4]	04
3. (a) Define continuous function with respect to some topology with example. [L2; CLO3]	03
(b) Draw the diagram and prove that the function $f(x) = x $ is continuous. [L5; CLO4]	04
(c) Define homeomorphic spaces and give examples. Also define topological property and give two examples which are not topological property. [L3; CLO4]	03
4. (a) Define metric space, usual metric and trivial metric. Give an example which is not metric space. [L3; CLO4]	03
(b) Let $C[0, 1]$ denote the collection of continuous functions on $[0, 1]$ and let $d(f, g) = \int_0^1 f(x) - g(x) dx$ and $d^*(f, g) = \sup\{ f(x) - g(x) : x \in [0, 1]\}$. Draw their graphs. Are they metric? [L1; CLO4]	04
(c) Define open sphere. Graphically explain three unusual spheres. [L3; CLO3]	03

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

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|----|-----|--|----|
| 5. | (a) | Define product space. Consider the topology $T = \{X, \phi, \{a\}, \{b, c\}\}$ on $X = \{a, b, c\}$ and the topology $T^* = \{Y, \phi, \{u\}\}$ on $Y = \{u, v\}$. Determine the subbase and base of the product topology $X \times Y$. [L3; CLO5] | 04 |
| | (b) | Prove that the usual topology on $R^2 = R \times R$ is the product topology. [L2; CLO5] | 03 |
| | (c) | Topologically draw the graph of unit square, annulus, torus and solid torus. [L4; CLO5] | 03 |
| 6. | (a) | Let $B = \{D_p : p \in Z \times Z\}$ where D_p has centre p and radius 1 and $1/2$ respectively. Will B cover R^2 ? Justify your answer. [L2; CLO6] | 03 |
| | (b) | Prove that continuous image of a compact set is compact. [L1; CLO6] | 03 |
| | (c) | State Heine-Borel theorem. Let A be any subset of a topological space X . Then prove that A is compact. [L5; CLO6] | 04 |
| 7. | (a) | State and Prove the Cantor's intersection theorem. [L5; CLO4] | 05 |
| | (b) | Show that a closed subspace of a normal space is a normal space. [L2; CLO4] | 05 |
| 8. | (a) | Define connected set and separated sets with figure. [L3; CLO7] | 03 |
| | (b) | Define path and homotopic path with figure. [L3; CLO8] | 03 |
| | (c) | Prove that $(0,1)$ interval is not compact. [L2; CLO8] | 04 |

***** THE END *****

Khulna University
Science, Engineering and Technology School
Discipline: Mathematics, Program: B.Sc. (Hons.)
Year: Fourth, Term: I Examination-2026; Session: 2024-2025
Course Code: 0541 12 Math 4111
Course Title: Differential Geometry
Total Marks: 60, Time: 03 Hours, Credit: 03



- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions **Marks**

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|----|-----|--|----|
| 1. | (a) | Define functions of class-r and equivalent paths. [L1; CLO1] | 02 |
| | (b) | Deduce the formula for the arc length of a curve between two points in vector form. [L4; CLO1] | 04 |
| | (c) | Find the arc length of the curve given as the intersection of the surfaces $x^2 + y^2 = a^2, z = \tan^{-1}\left(\frac{y}{x}\right)$, from the point $(a, 0, 0)$ to (x, y, z) . [L1; CLO1] | 04 |
| | | | |
| 2. | (a) | Define tangent line. [L1; CLO2] | 01 |
| | (b) | Show that $\frac{d\vec{r}}{ds}$ is unit tangent vector to a space curve at the point $p(\vec{r})$. [L4; CLO2] | 04 |
| | (c) | Obtain the direction cosines of the tangent to the curve of intersection of the surfaces $F_1(x, y, z) = 0, F_2(x, y, z) = 0$. [L5; CLO2] | 05 |
| | | | |
| 3. | (a) | Define principle normal and bi-normal of space curve. [L1; CLO3] | 02 |
| | (b) | Find the equation and the direction of the principle normal and bi-normal of space curve. [L1; CLO3] | 04 |
| | (c) | Find the inflexional tangents at (x_1, y_1, z_1) on the surface $y^2z = 4ax$. [L1; CLO3] | 04 |
| | | | |
| 4. | (a) | Define the osculating plane of a curve at a point. Find its equation in terms of $\vec{r}'(t)$ and $\vec{r}''(t)$. When does the osculating plane not exist? [L1; CLO2] | 03 |
| | (b) | Show that if a curve lies in a plane, its binormal vector $\vec{B}(t)$ is constant. Conversely, if $\vec{B}(t)$ is constant, prove the curve is planar. [L2; CLO2] | 03 |
| | (c) | Derive the Frenet-Serret formulas. [L6; CLO2] | 04 |

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

- | | | |
|----|--|----|
| 5. | (a) Define circular helix. [L1; CLO5] | 01 |
| | (b) Find the equation of a circular helix. [L1; CLO5] | 04 |
| | (c) Show that the curvature and torsion of the helix $x = a \cos \theta, y = a \sin \theta, z = a\theta \tan \alpha$ are $\frac{\cos^2 \alpha}{a}$ and $\pm \frac{\sin \alpha \cos \alpha}{a}$, respectively. [L4; CLO5] | 05 |
| 6. | (a) Define intrinsic equation. [L1; CLO6] | 02 |
| | (b) State and prove the Uniqueness theorem for space curves. [L5; CLO6] | 04 |
| | (c) Obtain the curvature κ_1 and torsion τ_1 of the involute. [L5; CLO6] | 04 |
| 7. | (a) Define the circle of curvature for a space curve at a point where curvature $\kappa > 0$. Explain why the circle of curvature lies in the osculating plane and has radius $\frac{1}{\kappa}$. Why is it unique? [L1; CLO1] | 03 |
| | (b) Define the spherical indicatrix of the binormal $B(s)$. Also, deduce the curvature and torsion of the spherical indicatrix of binormal. [L1; CLO2] | 03 |
| | (c) Find a unit normal vector to the surface $2xz^2 - 3xy - 4x = 7$ at the point $(1, -1, 2)$. [L1; CLO7] | 04 |
| 8. | (a) Explain that the first fundamental form is a quadratic form that measures arc length on the surface. Why is it called a "metric"? [L3; CLO4] | 03 |
| | (b) Compute the first fundamental form for the surface $r(u, v) = (u \cos v, u \sin v, u^2)$ at a general point (u, v) . [L3; CLO4] | 04 |
| | (c) Show that the curve bisecting the angles between the parametric curves are given by $Edu^2 - Gdv^2 = 0$. [L4; CLO8] | 03 |

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 Discipline: **Mathematics**, Program: **B.Sc. (Hons.)**
 Year: **Fourth**, Term: **I** Examination-2026; Session: **2024-2025**
 Course Code: **0541 12 Math 4103**
 Course Title: **Mathematical Modeling**
 Total Marks: **60**, Time: **03** Hours, Credit: **03**



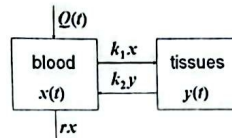
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Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

- Explain modeling and mathematical modeling in terms of dynamical systems. Discuss the different processes of mathematical modeling with appropriate flow chart. [L1; CLO1] **03**
 - How balance principle is preserved in mathematical modeling? Explain with your argument. [L2; CLO1] **04**
 - Incorporating all possible limitations of unbounded growth model, derive a logistic growth model for single species population. Solve the model with stability analysis. [L3; CLO2] **03**
- Suppose the population $p(t)$ is given by the exponential growth law. Growth rate r is positive. Let T be the amount of time required for the starting population p_0 to double. Then show that $T = \frac{\log_e 2}{r}$ which is independent of p_0 . [L2; CLO2] **06**
 - How mathematical model can be applied to kidney dialysis process? Describe and formulate an artificial kidney dialysis model and discuss the solution of the model. [L3; CLO2] **04**
- Explain pharmacokinetics. Describe and formulate a pharmacokinetic model using the following diagram. Solve the model with all possible equilibrium points and stability analysis. [L2; CLO3] **06**



- A symbiotic linear model can be formulated by the following governing equations. **04**

$$\frac{dx}{dt} = -x + 2y, \frac{dy}{dt} = 2x - y$$
 where $x(0) = 100$ and $y(0) = 300$. Check the nature of stability at $(0,0)$. Determine the populations $(x(t), y(t))$ at all future times t . What will be the populations after six months? When will the species y be eliminated? [L3; CLO3]
- State and explain two species Lotka-Volterra competition model. [L1; CLO4] **05**
 - What is traffic flow? What are the objectives of traffic flow modeling. Numerically solve the LWR (Lighthill, Whitham and Richards) traffic flow model $\frac{\partial \rho}{\partial t} + \frac{\partial(\rho v)}{\partial x} = 0$, $v = v(\rho)$. Given, Initial conditions: $\rho(x, 0) = \rho_0(x)$ and Boundary conditions: $\rho(0, t) = \alpha$, $\rho(L, t) = \beta$. [L4; CLO4] **05**

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

5. (a) Discuss the importance of discrete model. Construct an SIR type continuous model of disease dynamics. Convert this continuous model in the form of difference equation and solve with stability analysis. [L2; CLO5] **06**
 (b) Suppose "bugbox" is a simple computer **simulation** for structured population and it functions as a **biology lab for a virtual world**. For this "bugbox" model, let L_t be the number of larvae at time t , J_t be the number of juveniles at time t and A_t be the number of adults at time t . Formulate a structured population model and discuss the solution of the model with stability analysis. [L4; CLO5] **04**

6. Consider the pre-predator model with functional response $\frac{dx}{dt} = rx \left(1 - \frac{x}{k}\right) - bxy$, $\frac{dy}{dt} = cxy - dy$, where x is prey and y is predator. Examine the positivity and analyze its stability. [L3; CLO6] **10**

7. Discuss different *force of infections* in case of disease transmission. An SIRS epidemic model with nonlinear saturated incident rate can be constructed as $\frac{dS}{dt} = (1-p)A - \frac{\beta SI}{1+\alpha I} - dS + \gamma R$, $\frac{dI}{dt} = pA + \frac{\beta SI}{1+\alpha I} - (d+m)I$, $\frac{dR}{dt} = mI - (d+\gamma)R$, where $(1-p)A$ and pA are the constant recruitments of S and I individuals. Describe the model with a schematic diagram. Find the equilibrium points and test the stability at disease free equilibrium point. Explain basic reproduction number. [L5; CLO5] **10**

8. Consider two different types of foodstuffs, say, F_1 and F_2 . Assume that these foodstuffs contain vitamins V_1 , V_2 and V_3 respectively. Minimum daily requirements of these vitamins are 1mg of V_1 , 50mg of V_2 and 10mg of V_3 . Suppose that the foodstuffs F_1 contains 1mg of V_1 , 100mg of V_2 and 10mg of V_3 , whereas foodstuffs F_2 contains 1mg of V_1 , 10mg of V_2 and 100mg of V_3 . Cost of one unit of foodstuffs F_1 is \$1 and that of F_2 is \$1.5. Formulate a linear programming model to find the minimum cost diet that would supply the body at least minimum requirements of each vitamin. [L3; CLO6] **10**

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